

DURGESH SAKHARDANDE

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Experience

Community Dreams Foundation

June 2025 – June 2026

Software Development Engineer

Irving, TX

- **Engineered and maintained Phoenix Health Platform**, a full-stack production app (React/TypeScript, PostgreSQL via Supabase) for patients, owning the full lifecycle from feature design to deployment, including an end-to-end rebuild of the auth flow (session handling, token refresh, auth state sync)
- **Engineered an OCR/ETL pipeline** using Google Vision API and GPT-4o-mini processing **200+ medical documents/month**, extracting **25+ structured clinical biomarker fields** at an optimized cost of **~\$0.0004/report** through iterative prompt and token optimization
- **Conducted a full security audit and remediated** IDOR vulnerabilities, misconfigured CORS policies, JWT handling gaps, and PII/PHI exposure risks, bringing the platform into compliance with healthcare data handling standards
- **Designed and shipped an admin dashboard** with role-based access control (RBAC), a user invitation system, and role management, enabling non-technical staff to manage platform users without engineering intervention
- **Built a recommendation feature** on the Gemini API that translated structured biomarker output into personalized health insights surfaced in the patient-facing dashboard

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL

Frameworks: Spring Boot, FastAPI, Django (DRF), React, REST APIs, Microservices

Databases & Search: PostgreSQL, Elasticsearch, Redis

Cloud & DevOps: AWS (EC2, S3, IAM), Terraform, Docker, GitHub Actions, Linux

Messaging & Auth: RabbitMQ, OAuth2 (Keycloak), JWT

AI APIs & LLM Tooling: Gemini API, OpenAI (GPT-4o-mini), Google Vision API, Ollama, LangGraph

Projects

Multi-Agent Research Automation | *FastAPI, LangGraph, Gemini API, Python*

July 2026 – Present

- Built a **LangGraph** agent pipeline with a **PlannerAgent** that decomposes a query into sub-tasks using **schema-validated structured output**, then dispatches web retrieval (Tavily) and Gemini-based synthesis
- Hardened external API calls with **exponential backoff and retry**, and instrumented the pipeline to measure per-run latency and token cost against a single-agent baseline

AI Fitness Tracker | *Java, Spring Boot, RabbitMQ, Keycloak*

Aug 2025 – Oct 2025

- Built **4 independent Spring Boot services** communicating over **RabbitMQ**, exposing **8+ REST endpoints** behind an API Gateway with **OAuth2** (Keycloak) authorization and Eureka service discovery
- Diagnosed slow read paths with **query plan analysis** and resolved them through **composite indexing** and schema normalization, removing full table scans from the hot query path

Pantry to Plate | *Django (DRF), Elasticsearch, PostgreSQL, Redis, Docker*

Nov 2025 – Aug 2026

- Resolved **2.1M** free-text ingredient strings from a **231K-recipe** corpus to a canonical vocabulary, raising match coverage from **69% to 87%** via staged exact/alias/fuzzy matching
- Built coverage-based recipe ranking in **PostgreSQL** and **Elasticsearch**; verified backend equivalence for a **5x** speedup (170ms → 30ms), with **Redis** caching at **1.5ms**

Automated Cloud Deployment Platform | *AWS, Terraform, Docker, GitHub Actions*

Jan 2026 – Mar 2026

- Provisioned **AWS** infrastructure as code with **Terraform** across development and staging environments, and wired **GitHub Actions** pipelines that build, test, and publish Docker images on every commit

Education

Northeastern University

Sep 2023 – May 2025

Master of Science in Information Systems

Boston, MA

University of Mumbai

Aug 2018 – May 2022

Bachelor of Engineering in Electronics and Telecommunication

Mumbai, India

Relevant Coursework: Program Structure and Algorithms, Data Management and Database Design, Designing Advanced Data Architectures for Business Intelligence